eigenvalue is 0 and since repeated application of $\hat{a}^\dagger$ will increase the eigenvalue in steps of 1, all positive integer values can be reached.

The eigenvalues of the number operator $\hat{n}$ are thus all integers $n \geq 0$ and we denote the eigenstates by $|n\rangle$. These are also the energy eigenstates of the harmonic oscillator (because of (3.22)) and are often referred to as number states or *Fock states*. The energy eigenvalues are

$$E_n = \hbar\omega(n + \frac{1}{2}) \qquad \text{for } n = 0, 1, 2, \ldots. \tag{3.29}$$

The ground state $|0\rangle$ has energy $E_0 = \frac{1}{2}\hbar\omega$. We can think of n as the number of excitations in the oscillator. The ground state, which has no excitations is often referred to as the *vacuum state* (in particular for light). The annihilation and creation operators destroy and create excitations, respectively (hence the names). Since $\hat{a}|n\rangle$ and $\hat{a}^\dagger|n\rangle$ are eigenstates with eigenvalues $n-1$ and $n+1$, we must have $\hat{a}|n\rangle \propto |n-1\rangle$ and $\hat{a}^\dagger|n\rangle \propto |n+1\rangle$. We can find the factors of proportionality. Since

$$\langle n|\hat{a}^\dagger\hat{a}|n\rangle = \langle n|\hat{n}|n\rangle = n \tag{3.30}$$

and

$$\langle n|\hat{a}\hat{a}^\dagger|n\rangle = \langle n|(\hat{n} + [\hat{a}, \hat{a}^\dagger])|n\rangle = n + 1, \tag{3.31}$$

we must have (up to global phases which we can set to 1)

$$\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{3.32}$$

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle. \tag{3.33}$$

In particular, we have that $(\hat{a}^\dagger)^n|0\rangle = \sqrt{n(n-1)\cdots 1}|n\rangle = \sqrt{n!}|n\rangle$. The normalised eigenstates can therefore all be constructed from the ground state as

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle. \tag{3.34}$$

The Fock states form an orthonormal basis for the infinite-dimensional Hilbert space of the harmonic oscillator

$$\langle m|n\rangle = \delta_{mn}. \tag{3.35}$$

Any general pure state of a harmonic oscillator can be expressed in the Fock basis as

$$|\psi\rangle = \sum_{n=0}^{\infty} c_n|n\rangle. \tag{3.36}$$

3.1.3 Wavefunctions

In addition to the Fock basis representation, other representations of harmonic-oscillator states are also useful. In particular, it is common to perform measurements of $\hat{x}$ or $\hat{p}$ (or combinations of them), e.g. for light where they correspond quadratures of the electromagnetic field, which can be measured by homodyne detectors. Therefore, it is instructive to look at the wavefunctions of the Fock states.

Let us denote the $\hat{x}$-space wavefunction (c.f. Sec. 2.1.5) of the Fock state $|n\rangle$ by $\psi_n(x) = \langle x|n\rangle$. We start by finding the wavefunction for the ground state